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MZsearch: A Python-Based Compound Identification Tool for GC-MS and LC-MS/MS-Based Metabolomics

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Summary

A primary goal in metabolomics, the qualitative and quantitative study of metabolites, is to identify chemical compounds in a given sample. This often involves identifying and quantifying metabolic biomarkers for use in diagnosing, treating, and stratifying the risk of various diseases. A central tool for compound identification in metabolomics is mass spectrometry, which produces the mass spectrum for a given chemical compound. Each peak in the mass spectrum represents an ion fragment, with one axis representing mass to charge ratio (m/z) and the other axis representing relative intensity. A popular method used to identify an unknown chemical compound based on its mass spectrum is spectral library matching. This process involves computing a measure of similarity between the mass spectrum of the unknown chemical compound and each mass spectrum in a reference library. The unknown chemical compound is then identified as the one from the reference library whose mass spectrum is most similar to that of the unknown compound. We present MZsearch, a command-line tool written in Python that implements spectral library matching. The MZsearch tool offers a range of spectrum preprocessing transformations and similarity measures, and it is capable of analyzing data generated from gas chromatography-mass spectrometry (GC-MS) and liquid chromatography-mass spectrometry (LC-MS/MS).

Statement of need

There exist several open-source software tools for working with mass spectrometry data. The Python package matchms allows one to perform spectral library matching for both GC-MS and LC-MS/MS and provides an application programming interface (API) for users to define their own spectrum preprocessing transformations and similarity measures (Huber et al., 2020). The pyOpenMS tool, which is a Python wrapper of the OpenMS tool, is for analyzing LC-MS/MS data and provides common spectrum preprocessing transformations such as filtering based on m/z and intensity values (Röst et al., 2014, 2016). The Python package spectrum_utils provides functionality for preprocessing and visualizing mass spectrometry data (Bittremieux, 2020). The R package metID is a tool for LC-MS/MS-based compound identification with a focus on allowing users to combine their in-house database(s) with public databases (Shen et al., 2022). The R Shiny package ShinyMetID provides users six similarity measures (Cosine, Weighted Cosine, Stein & Scott, Discrete Fourier Transformation, Discrete Wavelet Transformation, and Semi-Partial Correlation) and a graphical user interface (GUI) to perform spectral library matching to identify chemical compounds from GC-MS data (Oh et al., 2023).

In many GC-MS experiments, the resulting mass spectra often have m/z values reported as integers due to the use of nominal mass resolution instruments. In contrast, LC-MS/MS frequently employs high-resolution mass spectrometers that provide m/z values with multiple

decimal places. This necessitates several spectrum preprocessing steps unique to LC-MS/MS data analysis, namely centroiding (i.e., merging peaks that are 'close' with respect to their m/z values) and matching (i.e., aligning the m/z values so that the query spectrum and reference spectrum have the same length). In addition to these canonical spectrum preprocessing transformations for LC-MS/MS data, weight factor transformations and low-entropy transformations have been proposed to improve the performance of compound identification for both GC-MS and LC-MS/MS data (Dlugas et al., 2024; Kim et al., 2012; Li et al., 2021). The Shannon Entropy Similarity Measure has been shown to outperform the Cosine Similarity Measure with respect to LC-MS/MS data (Li et al., 2021). A generalization of the Shannon Entropy Similarity Measure, the Tsallis Entropy Similarity Measure, slightly outperforms the Shannon Entropy Similarity Measure in both GC-MS and LC-MS/MS data (Dlugas et al., 2024). This recent study has further demonstrated that the order of preprocessing transformations is also critical to achieving higher accuracy in compound identification. However, to our knowledge, there is no tool that includes the recently-developed, high-performance entropy-based similarity measures or considers the order of the spectrum preprocessing transformations, which are important for effective metabolomics studies.

To address the lack of spectral library matching software that considers both the order of spectrum preprocessing steps and novel entropy-based similarity measures, MZsearch was developed. The developed MZsearch is a command-line tool for performing spectral library matching on either GC-MS or LC-MS/MS data. It allows users to construct their own spectrum preprocessing order using spectrum preprocessing transformations such as weight factor and low-entropy transformations. Additionally, the MZsearch tool includes a novel entropy-based similarity measure, the Rényi Entropy Similarity Measure, enabling users to choose among four similarity measures: the commonly-used Cosine Similarity Measure (Stein & Scott, 1994), the Shannon Entropy Similarity Measure (Li et al., 2021), the recently-developed Entropy Similarity Measure (Dlugas et al., 2024), and the novel Rényi Entropy Similarity Measure. Table 1 compares MZsearch with other similar software.

Table 1. Comparison of MZsearch and existing tools.

Package	Program language	User interface	LC-MS/MS compatible	GC-MS compatible	Similarity measures	Preprocessing/Transformation(s)
`MZsearch`	Python	command line; Python coding	yes	yes	cosine; Shannon; Rényi; Tsallis	weight factor; low-entropy; filtering on m/z and intensity values; noise removal
`matchms` (Huber et al., 2020)	Python	Python coding	yes	no	cosine-based; Dice; Jaccard; custom	filtering on m/z and intensity values; noise removal; custom

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Package	Program language	User interface	LC-MS/MS compatible	GC-MS compatible	Similarity measures	Preprocessing/Transformation(s)
`OpenMS` (Röst et al., 2014, 2016)	C++/Python	C++/Python coding	yes	no	cosine	smoothing; filtering; other functionality not related to compound identification
`spectrum_utils` (Bittremieux, 2020)	Python	Python coding	yes	no	N/A	precursor & noise peak removal; intensity filtering; intensity scaling
`metID` (Shen et al., 2022)	R	R coding	yes	no	cosine	noise removal
`ShinyMetID` (Oh et al., 2023)	R	R/Shiny GUI	no	yes	cosine; weighted cosine; Stein & Scott; discrete Fourier transform; discrete wavelet transform; semi-partial correlation	weight factor

Functionality

Spectrum Preprocessing Transformations

Functionality implementing the following spectrum preprocessing transformations is offered in MZsearch (more details found in documentation:):

- Filtering based on mz-values and intensities
- Weight Factor Transformation
- Low-Entropy Transformation
- Noise Removal
- Centroiding (only applicable to LC-MS/MS data)
- Matching (only applicable to LC-MS/MS data)

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- Normalization (only applicable to entropy-based similarity measures)

The flowchart in Figure 1 depicts the overall workflow of MZsearch.

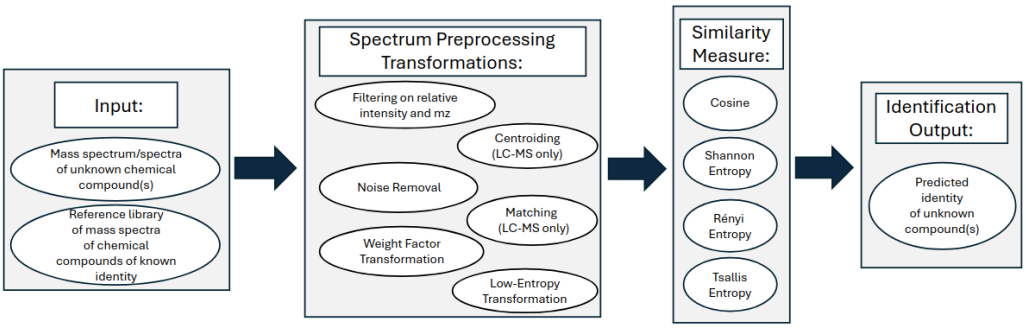

Figure 1: Workflow of MZsearch.

Similarity Measures

Given a pair of processed spectra intensities $I = (a_1, a_2, ..., a_n), J = (b_1, b_2, ..., b_n) \in \mathbb{R}^n$ with $0 \leq a_i, b_i \leq 1$ for all $i \in \{1, 2, ..., n\}$ and $\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} b_i = 1$, MZsearch provides functionality for computing the following similarity measures:

- Cosine Similarity Measure:

$$S_{Cosine}(I, J) = \frac{I \circ J}{|I|_2 \cdot |J|_2}$$

  where multiplication in the numerator refers to the dot product $I \circ J = a_1 b_1 + a_2 b_2 + ... + a_n b_n$ and multiplication in the denominator refers to multiplication of the L^2-norm of I and J, $|I|_2 = \sqrt{a_1^2 + a_2^2 + ... + a_n^2}, |J|_2 = \sqrt{b_1^2 + b_2^2 + ... + b_n^2}$.

- Shannon Entropy Similarity Measure:

$$S_{Shannon}(I, J) = 1 - \frac{2 \cdot H_{Shannon}\left(\frac{I+J}{2}\right) - H_{Shannon}(I) - H_{Shannon}(J)}{ln(4)},$$

$$H_{Shannon}(I) = -\sum_{i=1}^{n} a_i \cdot ln(a_i)$$

- Tsallis Entropy Similarity Measure (Havrda & Charvát, 1967; Tsallis, 1988):

$$S_{Tsallis}(I, J, q) = 1 - \frac{2 \times H_{Tsallis}(I/2 + J/2, q) - H_{Tsallis}(I, q) - H_{Tsallis}(J, q)}{N_{Tsallis}(I, J, q)},$$

$$N_{Tsallis}(I, J, q) := \frac{\sum_{i=1}^{n} \left(2\left(\frac{a_i}{2}\right)^q + 2\left(\frac{b_i}{2}\right)^q - a_i^q - b_i^q\right)}{1 - q},$$

$$H_{Tsallis}(I, q) = \frac{\left(\sum_{i=1}^{n} a_i^q\right) - 1}{1 - q},$$

$$q \neq 1, \ q > 0$$

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- Rényi Entropy Similarity Measure:

$$S_{Renyi}(I,J,q) = 1 - \frac{2 \times H_{Renyi}(I/2 + J/2, q) - H_{Renyi}(I,q) - H_{Renyi}(J,q)}{N_{Renyi}(I,J,q)},$$

$$N_{Renyi}(I,J,q) := \left(\frac{1}{1-q}\right)\left(2 \times ln\left(\sum_i (a_i/2)^q + \sum_j (b_j/2)^q\right) - ln(\sum_i a_i^q) - ln(\sum_i b_i^q)\right)$$

$$H_{Renyi}(I,q) = \frac{1}{1-q} ln(\sum_{i=1}^{n} a_i^q),$$

$$q \neq 1,\ q > 0$$

Usage

Compound identification occurs during preprocessing or as an independent procedure. In preprocessing, it follows peak detection and should be performed before cross-sample alignment if based on compound names; otherwise, it can be delayed until before pathway analysis. As an independent procedure, it can be conducted ad hoc or post hoc to confirm or identify specific peaks in single or multiple samples. MZsearch performs compound identification and has three main capabilities: (i) converting the raw data to the necessary format for spectral library matching, (ii) running spectral library matching to identify compounds based on their mass spectrometry data, and (iii) plotting a query spectrum vs. a reference spectrum before and after preprocessing transformations. These tasks are implemented separately for (i) GC-MS and (ii) LC-MS/MS data due to the different spectrum preprocessing transformations. To see all parameters for any of the three main scripts (build_library.py, spec_lib_matching.py, plot_spectra.py), see the documentation at https://github.com/hdlugas/MZsearch.

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Author Contributions

S.K. conceived the study and contributed to the overall study design and interpretation. H.D., X.Z., and S.K. collaborated on discussing the design, features, and structures of the developed software package. H.D. implemented the initial idea, further expanded and refined it, and was responsible for coding the developed software package. H.D., X.Z., and S.K. reviewed and edited the manuscript. Both H.D. and S.K. contributed equally to the manuscript preparation, including drafting and revising the manuscript.

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